

Machine Learning — Linear Regression

Numerical Question Paper

Simple • Medium • Hard

Simple Level

Question 1

Given the data points (1, 2), (2, 4), and (3, 6), calculate the slope m of the regression line.

Question 2

Using the same data points (1, 2), (2, 4), and (3, 6), calculate the intercept b .

Question 3

For the linear regression model below, calculate the predicted value when $x = 5$.

$$\hat{y} = 2x + 1$$

Question 4

Actual values are $y = [3, 5, 7]$ and predicted values are $\hat{y} = [2, 6, 8]$. Calculate the Mean Squared Error (MSE).

$$MSE = (1/n) \sum (y_i - \hat{y}_i)^2$$

Medium Level

Question 5

Given $x = [1, 2, 3, 4]$ and $y = [3, 5, 7, 9]$, calculate the regression equation $\hat{y} = mx + b$.

Question 6

Given the feature matrix X and target vector y below, calculate θ using the closed-form normal equation.

Feature matrix X :

1	1
1	2
1	3

Target vector y :

2
4
6

$$\theta = (X^T X)^{-1} X^T y$$

Question 7

For the model $\hat{y} = 2x + 1$ and data points (1, 4), (2, 5), and (3, 8), calculate the MSE.

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Question 8

Perform one gradient descent update using $x = 2$, $y = 6$, initial $w = 1$, $b = 0$, and learning rate $\alpha = 0.1$. Use the cost function shown below.

$$\hat{y} = wx + b$$

$$J = \frac{1}{2}(\hat{y} - y)^2$$

$$\partial J / \partial w = (\hat{y} - y)x \quad \partial J / \partial b = \hat{y} - y$$

Hard Level

Question 9

Using batch gradient descent, perform one iteration for data points (1, 3), (2, 5), and (3, 7). Start with $w = 0$, $b = 0$ and learning rate $\alpha = 0.1$. Calculate the updated w and b .

$$J = (1/2m) \sum (\hat{y}_i - y_i)^2$$

Question 10

For $x = [1, 2, 3]$ and $y = [2, 4, 5]$, calculate the closed-form slope and intercept, then predict y when $x = 4$.

$$\theta = (X^T X)^{-1} X^T y$$

Question 11

Given the feature matrix X and target vector y below, calculate θ using the closed-form solution.

Feature matrix X:

1	1
1	2
1	4

Target vector y:

2
3
7

$$\theta = (X^T X)^{-1} X^T y$$

Question 12

A model starts with $w = 1$ and $b = 1$. For $x = 3$ and $y = 10$, perform one gradient descent step with learning rate $\alpha = 0.05$. Find the updated w and b .

$$\hat{y} = wx + b$$

$$J = \frac{1}{2}(\hat{y} - y)^2$$

$$\partial J / \partial w = (\hat{y} - y)x \quad \partial J / \partial b = \hat{y} - y$$